This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Part 1: Data

- ☐ This paper does not involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).
- ☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

The paper uses three sources of data:

- Randomly generated data
- The publicly available Stan examples
- A publicly available textbook dataset for MrP

Availability

- ☒ Data **are** publicly available.
- ☐ Data **cannot be made** publicly available.

If the data are publicly available, see the *Publicly available data* section. Otherwise, see the *Non-publicly available data* section, below.

Publicly available data

- ☒ Data are available online at:

MrP data: https://github.com/JuanLopezMartin/MRPCaseStudy/tree/master/data_public/chapter1/data
Stan data: https://github.com/stan-dev/example-models

- ☒ Data are available as part of the paper's supplementary material.
- ☐ Data are publicly available by request, following the process described here:
- ☐ Data are or will be made available through some other mechanism, described here:

Non-publicly available data

Description

File format(s)

- ☐ CSV or other plain text.
- ☐ Software-specific binary format (.Rda, Python pickle, etc.): pkcle
- ☐ Standardized binary format (e.g., netCDF, HDF5, etc.):
- ☐ Other (please specify):

Data dictionary

- ☐ Provided by authors in the following file(s):
- ☐ Data file(s) is(are) self-describing (e.g., netCDF files)
- ☒ Available at the following URL:

We refer readers to the original sources.

MrP data: https://github.com/JuanLopezMartin/MRPCaseStudy/tree/master/data_public/chapter1/data
Stan data: https://github.com/stan-dev/example-models

Additional Information (optional)

Part 2: Code

Abstract

Our three analyses are pipelines written in R, occasionally with bash wrappers for parallelism with slrum. After downloading the data from its sources, follow the README instructions to run the R scripts in order, or use the provided makefiles.

The pipelines rely on a number of R packages that we provide, which provide re-usable versions of common functionality.

Description

Code format(s)

- ☒ Script files
  - ☒ R
  - ☐ Python
  - ☐ Matlab
  - ☐ Other:
- ☒ Package
  - ☒ R
  - ☐ Python
  - ☐ MATLAB toolbox
  - ☐ Other:
- ☐ Reproducible report
  - ☐ R Markdown
  - ☐ Jupyter notebook
  - ☐ Other:
- ☒ Shell script
- ☐ Other (please specify):

Supporting software requirements

@Article{, title = {{brms}: An {R} Package for {Bayesian} Multilevel Models Using {Stan}}, author = {Paul-Christian Bürkner}, journal = {Journal of Statistical Software}, year = {2017}, volume = {80}, number = {1}, pages = {1–28}, doi = {10.18637/jss.v080.i01}, encoding = {UTF-8}, }

@Misc{, title = {rstanarm: {Bayesian} applied regression modeling via {Stan}.}, author = {Ben Goodrich and Jonah Gabry and Imad Ali and Sam Brilleman}, note = {R package version 2.32.1}, year = {2024}, url = {https://mc-stan.org/rstanarm/}, }

@Misc{, title = {{RStan}: the {R} interface to {Stan}}, author = {{Stan Development Team}}, note = {R package version 2.32.7}, year = {2025}, url = {https://mc-stan.org/}, }

Version of primary software used platform x86_64-pc-linux-gnu
arch x86_64
os linux-gnu
system x86_64, linux-gnu
status

major 4
minor 5.3
year 2026
month 03
day 11
svn rev 89597
language R
version.string R version 4.5.3 (2026-03-11) nickname Reassured Reassurer

Libraries and dependencies used by the code

`brms`	`2.22.0`
`broom`	`1.0.8`
`broom.mixed`	`0.2.9.6`
`devtools`	`2.4.5`
`doParallel`	`1.0.17`
`dplyr`	`1.1.4`
`ggplot2`	`3.5.2`
`gridExtra`	`2.3`
`jsonlite`	`2.0.0`
`latex2exp`	`0.9.6`
`lme4`	`1.1.37`
`mcmcse`	`1.5.0`
`mvtnorm`	`1.3.3`
`numDeriv`	`2016.8.1.1`
`optparse`	`1.7.5`
`reshape2`	`1.4.4`
`rstan`	`2.32.7`
`rstanarm`	`2.32.1`
`testthat`	`3.2.3`
`tidybayes`	`3.0.7`
`tidyverse`	`2.0.0`

Supporting system/hardware requirements (optional)

There are no special hardware requirements.

Parallelization used

- ☒ No parallel code used
- ☐ Multi-core parallelization on a single machine/node
  - Number of cores used:
- ☐ Multi-machine/multi-node parallelization
  - Number of nodes and cores used:

In order to speed up the many MCMC runs, we provide optional slurm versions for parallelism. But the code can be run serially as well.

License

- ☒ MIT License (default)
- ☐ BSD
- ☐ GPL v3.0
- ☐ Creative Commons
- ☐ Other: (please specify)

Additional information (optional)

The code is available as part of the paper submission, as well as at the Github repository

`https://github.com/rgiordan/BayesIJPaper`

Part 3: Reproducibility workflow

Scope

The provided workflow reproduces:

- ☒ Any numbers provided in text in the paper
- ☒ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))
- ☒ All tables and figures in the paper
- ☐ Selected tables and figures in the paper, as explained and justified below:

Workflow

The workflow is documented in the README and makefiles.

Location

The workflow is available:

- ☒ As part of the paper's supplementary material.
- ☒ In this Git repository: https://github.com/rgiordan/BayesIJPaper
- ☐ Other (please specify):

Format(s)

- ☐ Single master code file
- ☐ Wrapper (shell) script(s)
- ☐ Self-contained R Markdown file, Jupyter notebook, or other literate programming approach
- ☒ Text file (e.g., a readme-style file) that documents workflow
- ☒ Makefile
- ☐ Other (more detail in *Instructions* below)

Instructions

Follow the README and makefile instructions.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:

- ☐ < 1 minute
- ☐ 1-10 minutes
- ☐ 10-60 minutes
- ☐ 1-8 hours
- ☐ > 8 hours
- ☒ Not feasible to run on a desktop machine, as described here:

The paper verifies frequentist sampling accuracy by running a very large number of posteriors on bootstrapped or resampled datsets. These analyses are set up to run on slurm, but could in principle be run on a desktop.

Additional information (optional)

Notes (optional)